Lesson 14: Stochastic Approximation Noise Decomposition

AI/ML Mathematics

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1 Motivation and Overview

This is the capstone lesson of the Measure Theory track (Lessons 1–14). Over the preceding thirteen lessons we assembled a complete probability toolkit: sigma-algebras and measures (Lessons 1–2), measurable functions and expectation (Lessons 3–5), conditional expectation and variance (Lessons 6–7), filtrations and martingale difference sequences (Lesson 8), concentration inequalities (Lesson 9), modes of convergence and Borel–Cantelli (Lesson 10), limit theorems (Lesson 11), Markov chains (Lesson 12), and martingale convergence theory including the Robbins–Siegmund theorem (Lesson 13).

The purpose of this lesson is twofold. First, we prove the *TD noise decomposition theorem*, which shows that the noise in a temporal-difference update is a martingale difference sequence with bounded conditional variance. This single result is the bridge between measure theory and the stochastic approximation convergence theorems developed in the Optimization track (Lessons 31–32). Second, we provide a comprehensive summary of every key result from Lessons 1–14, organized by its role in RL theory.

Why This Matters for RL

- **TD(0), SARSA, Q-learning convergence:** Every convergence proof for a tabular RL algorithm reduces the update to Robbins–Monro form: $\theta_{n+1} = \theta_n + \alpha_n[h(\theta_n) + w_n]$. The TD noise decomposition proves that $\{w_n\}$ is an MDS with the right variance bound, which is the main probabilistic ingredient.
- **Two-timescale actor-critic:** Actor-critic algorithms run two coupled SA recursions. Each recursion needs its own MDS noise decomposition; the structure of the proof is identical to the one given here.
- **From theory to practice:** Understanding *why* the noise is an MDS (and not just "mean zero") clarifies which algorithm modifications preserve convergence guarantees and which break them.

Prerequisites. All of Lessons 1–13, especially: conditional expectation and the tower property (Lesson 6), martingale difference sequences (Lesson 8), and the Robbins–Siegmund theorem (Lesson 13).

Learning objectives.

- State and prove the TD noise decomposition theorem.
- Verify the MDS and bounded-variance conditions for TD(0), Q-learning, and expected SARSA.
- Connect the noise decomposition to the Robbins–Monro framework and state the resulting convergence theorem.
- Summarize the role of every key measure-theoretic result in RL convergence proofs.

2 References

- Borkar [2008], Chapter 2: Stochastic approximation framework and noise conditions.
- Sutton and Barto [2018], Chapters 6 and 12: TD learning algorithms and eligibility traces.
- Szepesvári [2010], Chapter 4: Convergence of TD and Q-learning with detailed noise analysis.
- Bertsekas [2012], Chapter 6: Approximate dynamic programming and stochastic approximation convergence.
- Williams [1991], Chapters 10–12: Martingales and conditional expectation—the probabilistic backbone.

3 The MDP Framework and Bellman Operators

We fix the notation used throughout this lesson. A *finite Markov Decision Process* (MDP) is a tuple $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ where $\mathcal{S}$ is a finite state space, $\mathcal{A}$ is a finite action space, $P(s'|s,a) = \mathbb{P}(S_{t+1} = s' \mid S_t = s, A_t = a)$ is the transition kernel, $r : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}$ is a bounded reward function with $|r(s,a,s')| \le R_{\max}$, and $\gamma \in [0,1)$ is the discount factor.

Definition 3.1 (Stationary Policy)**.** A stationary (randomized) policy is a mapping $\pi : \mathcal{S} \to \Delta(\mathcal{A})$, where $\pi(a|s)$ denotes the probability of selecting action a in state s.

Definition 3.2 (Value Function)**.** The state-value function under policy π is

$$V^\pi(s) = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t R_{t+1} \,\middle|\, S_0 = s\right], \qquad s \in \mathcal{S}. \tag{1}$$

The action-value function is $Q^\pi(s,a) = \mathbb{E}_\pi[\sum_{t=0}^{\infty} \gamma^t R_{t+1} \mid S_0 = s, A_0 = a]$.

Definition 3.3 (Bellman Operator)**.** The Bellman operator $\mathcal{T}^\pi : \mathbb{R}^{|\mathcal{S}|} \to \mathbb{R}^{|\mathcal{S}|}$ is defined by

$$(\mathcal{T}^\pi V)(s) = \sum_{a \in \mathcal{A}} \pi(a|s) \sum_{s' \in \mathcal{S}} P(s'|s,a)\big[r(s,a,s') + \gamma V(s')\big]. \tag{2}$$

The value function V^π is the unique fixed point of $\mathcal{T}^\pi$.

Remark 3.4. The operator $\mathcal{T}^\pi$ is a γ-contraction in the supremum norm: $\|\mathcal{T}^\pi V - \mathcal{T}^\pi W\|_\infty \le \gamma \|V - W\|_\infty$. This is proved using the linearity in V and the fact that $\sum_a \pi(a|s) \sum_{s'} P(s'|s,a) = 1$. The Banach fixed-point theorem (Lesson 22) then guarantees existence and uniqueness of V^π.

4 The TD(0) Update and Its Noise Structure

We now derive the noise structure of the simplest temporal-difference algorithm: tabular TD(0) for policy evaluation.

4.1 The TD(0) Update Rule

An agent follows policy π and observes a trajectory $S_0, A_0, R_1, S_1, A_1, R_2, \ldots$ The TD(0) update at time t modifies the value estimate at the visited state:

$$V_{t+1}(S_t) = V_t(S_t) + \alpha_t(S_t)\big[R_{t+1} + \gamma V_t(S_{t+1}) - V_t(S_t)\big], \tag{3}$$

where $\alpha_t(s) \ge 0$ is the step size applied to state s at the t-th visit, and $V_{t+1}(s) = V_t(s)$ for all $s \ne S_t$.

Definition 4.1 (TD Error)**.** The temporal-difference error at time t is

$$\delta_t = R_{t+1} + \gamma V_t(S_{t+1}) - V_t(S_t). \tag{4}$$

4.2 Rewriting TD(0) in Robbins–Monro Form

To apply stochastic approximation theory, we rewrite (3) by adding and subtracting $(\mathcal{T}^\pi V_t)(S_t)$:

$$V_{t+1}(S_t) = V_t(S_t) + \alpha_t(S_t)\big[(\mathcal{T}^\pi V_t)(S_t) - V_t(S_t) + w_t(S_t)\big], \tag{5}$$

where the *noise term* is

$$w_t(s) = R_{t+1} + \gamma V_t(S_{t+1}) - (\mathcal{T}^\pi V_t)(s). \tag{6}$$

The term $h(V_t)(s) = (\mathcal{T}^\pi V_t)(s) - V_t(s)$ is the *mean field*: it equals zero exactly when $V_t = V^\pi$. The noise $w_t(s)$ captures the randomness from the single-sample transition $(S_t, A_t, R_{t+1}, S_{t+1})$.

4.3 The Filtration

Definition 4.2 (Natural Filtration of the MDP Trajectory)**.** Define the filtration $\{\mathcal{F}_t\}_{t\geq 0}$ by

$$\mathcal{F}_t = \sigma\big(S_0, A_0, R_1, S_1, \ldots, S_{t-1}, A_{t-1}, R_t, S_t\big). \tag{7}$$

Crucially, V_t and S_t are $\mathcal{F}_t$-measurable, while A_t, R_{t+1}, and S_{t+1} are $\mathcal{F}_{t+1}$-measurable but *not* $\mathcal{F}_t$-measurable.

5 The TD Noise Decomposition Theorem

The following theorem is the main result of this lesson. It establishes the two conditions—zero conditional mean and bounded conditional variance—that connect the TD noise to the Robbins–Monro framework.

Theorem 5.1 (TD Noise Decomposition)**.** *Let $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ be a finite MDP with $|r(s,a,s')| \leq R_{\max}$. Let π be a stationary policy, and let $\{V_t\}$ be the sequence of value estimates produced by TD(0) with the noise $w_t(s)$ defined in (6). Then for all $t \geq 0$ and all $s \in \mathcal{S}$:*

1. ***(Martingale difference property)*** $\mathbb{E}[w_t(s) \mid \mathcal{F}_t] = 0$.
2. ***(Bounded conditional variance)*** *There exists a constant $C > 0$ (depending only on $R_{\max}$, γ, and $|\mathcal{S}|$) such that*

$$\mathbb{E}\big[w_t(s)^2 \mid \mathcal{F}_t\big] \leq C\big(1 + \|V_t\|_\infty^2\big). \tag{8}$$

Proof. **Part 1 (MDS property).** Conditioned on $\mathcal{F}_t$, the quantities V_t and $S_t = s$ are known constants. The randomness comes from the action $A_t \sim \pi(\cdot|s)$ and the next state $S_{t+1} \sim P(\cdot|s, A_t)$. Compute:

$$\mathbb{E}[w_t(s) \mid \mathcal{F}_t] = \mathbb{E}\big[R_{t+1} + \gamma V_t(S_{t+1}) \mid \mathcal{F}_t\big] - (\mathcal{T}^\pi V_t)(s). \tag{9}$$

For the first term, we apply the law of total expectation (tower property, Lesson 6) by conditioning on A_t:

$$\begin{aligned}
&\mathbb{E}\big[R_{t+1} + \gamma V_t(S_{t+1}) \mid \mathcal{F}_t\big] \\
&\quad = \sum_{a\in\mathcal{A}} \pi(a|s)\, \mathbb{E}\big[R_{t+1} + \gamma V_t(S_{t+1}) \mid S_t = s,\, A_t = a\big] \\
&\quad = \sum_{a\in\mathcal{A}} \pi(a|s) \sum_{s'\in\mathcal{S}} P(s'|s,a)\big[r(s,a,s') + \gamma V_t(s')\big] \\
&\quad = (\mathcal{T}^\pi V_t)(s).
\end{aligned} \tag{10}$$

In the second equality we used the Markov property: conditioned on (S_t, A_t), the pair (R_{t+1}, S_{t+1}) is independent of all earlier history, so conditioning on $\mathcal{F}_t$ reduces to conditioning on (S_t, A_t). Substituting (10) into (9):

$$\mathbb{E}[w_t(s) \mid \mathcal{F}_t] = (\mathcal{T}^\pi V_t)(s) - (\mathcal{T}^\pi V_t)(s) = 0.$$

Part 2 (Bounded conditional variance). Since $|r(s,a,s')| \leq R_{\max}$ and $|V_t(s')| \leq \|V_t\|_\infty$ for all s', we bound each component of $w_t(s)$:

$$\begin{aligned}
|R_{t+1}| &\leq R_{\max}, \\
|\gamma V_t(S_{t+1})| &\leq \gamma \|V_t\|_\infty, \\
|(\mathcal{T}^\pi V_t)(s)| &\leq \sum_a \pi(a|s) \sum_{s'} P(s'|s,a)\big[R_{\max} + \gamma\|V_t\|_\infty\big] = R_{\max} + \gamma\|V_t\|_\infty.
\end{aligned} \tag{11}$$

Therefore

$$\begin{aligned}|w_t(s)| &\le |R_{t+1}| + \gamma|V_t(S_{t+1})| + |(\mathcal{T}^\pi V_t)(s)| \\ &\le 2R_{\max} + 2\gamma\|V_t\|_\infty. \end{aligned} \tag{12}$$

Squaring and using $(a+b)^2 \le 2a^2 + 2b^2$:

$$\begin{aligned} w_t(s)^2 &\le (2R_{\max} + 2\gamma\|V_t\|_\infty)^2 \\ &\le 2(2R_{\max})^2 + 2(2\gamma\|V_t\|_\infty)^2 \\ &= 8R_{\max}^2 + 8\gamma^2\|V_t\|_\infty^2. \end{aligned} \tag{13}$$

Setting $C = 8\max(R_{\max}^2, \gamma^2)$, we obtain

$$\mathbb{E}\big[w_t(s)^2 \mid \mathcal{F}_t\big] \le C\big(1 + \|V_t\|_\infty^2\big).$$

□

TD Noise is a Martingale Difference Sequence

The noise $w_t(s) = R_{t+1} + \gamma V_t(S_{t+1}) - (\mathcal{T}^\pi V_t)(s)$ satisfies:

1. $\mathbb{E}[w_t(s) \mid \mathcal{F}_t] = 0$ (MDS property),
2. $\mathbb{E}[w_t(s)^2 \mid \mathcal{F}_t] \le C(1 + \|V_t\|_\infty^2)$ (bounded variance).

These are precisely the noise conditions required by the Robbins–Monro theorem (Lesson 31) and the ODE method (Lesson 32) to guarantee almost-sure convergence.

Remark 5.2 (Role of the Markov Property). The key step in Part 1 is that conditioning on the full history $\mathcal{F}_t$ reduces to conditioning on (S_t, A_t). This is the *Markov property* of the MDP. Without it—for example, in a POMDP where the agent cannot observe the true state—the noise would depend on the entire history and the MDS property would generally fail.

6 Extensions to Other Algorithms

The noise decomposition argument of Theorem 5.1 extends to other tabular RL algorithms with minor modifications.

6.1 Q-Learning

In Q-learning, the update target replaces the policy average with the greedy action:

$$Q_{t+1}(S_t, A_t) = Q_t(S_t, A_t) + \alpha_t\big[R_{t+1} + \gamma \max_{a'} Q_t(S_{t+1}, a') - Q_t(S_t, A_t)\big]. \tag{14}$$

The Bellman optimality operator is $(\mathcal{T}^* Q)(s,a) = \sum_{s'} P(s'|s,a)[r(s,a,s') + \gamma \max_{a'} Q(s',a')]$, and the noise is

$$w_t^Q(s,a) = R_{t+1} + \gamma \max_{a'} Q_t(S_{t+1}, a') - (\mathcal{T}^* Q_t)(s,a). \tag{15}$$

Proposition 6.1 (Q-Learning Noise Decomposition)**.** *Under the same setup as Theorem 5.1, the Q-learning noise $w_t^Q(s,a)$ satisfies:*

1. $\mathbb{E}[w_t^Q(s,a) \mid \mathcal{F}_t] = 0$,
2. $\mathbb{E}[w_t^Q(s,a)^2 \mid \mathcal{F}_t] \le C'(1 + \|Q_t\|_\infty^2)$ *for a constant C' depending only on $R_{\max}$ and γ.*

Proof. The proof is essentially identical to that of Theorem 5.1. For Part 1, conditioned on $\mathcal{F}_t$ (which includes Q_t and $S_t = s$, $A_t = a$), the next-state transition $S_{t+1} \sim P(\cdot|s,a)$ and reward R_{t+1} are independent of all prior history by the Markov property. Then

$$\begin{aligned} \mathbb{E}\big[R_{t+1} + \gamma \max_{a'} Q_t(S_{t+1}, a') \mid \mathcal{F}_t\big] &= \sum_{s'} P(s'|s,a)\big[r(s,a,s') + \gamma \max_{a'} Q_t(s',a')\big] \\ &= (\mathcal{T}^* Q_t)(s,a), \end{aligned} \tag{16}$$

giving $\mathbb{E}[w_t^Q \mid \mathcal{F}_t] = 0$. Part 2 follows from $|R_{t+1}| \le R_{\max}$, $|\max_{a'} Q_t(s', a')| \le \|Q_t\|_\infty$, and the same triangle-inequality and squaring argument. □

6.2 Expected SARSA

Expected SARSA replaces the sample next-action with its expectation under the behavior policy b:

$$V_{t+1}(S_t) = V_t(S_t) + \alpha_t\big[R_{t+1} + \gamma \textstyle\sum_{a'} b(a'|S_{t+1}) Q_t(S_{t+1}, a') - Q_t(S_t, A_t)\big]. \tag{17}$$

The noise decomposition again yields an MDS with bounded conditional variance. The key observation is that $\sum_{a'} b(a'|S_{t+1}) Q_t(S_{t+1}, a')$ is a bounded function of S_{t+1} (given Q_t), so the same argument applies.

When the MDS Property Fails

The MDS property relies on:

1. **Markov transitions:** If the environment is non-Markovian (e.g., a POMDP), the conditional expectation given the full history does not simplify.
2. **Fixed target network:** If V_t is updated between computing the target and the current estimate (as in some batched implementations), the cancellation in Part 1 may not hold exactly.
3. **Bounded rewards:** If rewards are unbounded, Part 2 fails. Extensions require sub-Gaussian or sub-exponential tail conditions.

7 Convergence via the Robbins–Monro Framework

We now state how the noise decomposition feeds into the convergence machinery.

7.1 The Robbins–Monro Template

The Robbins–Monro stochastic approximation theorem (Lesson 31) considers iterations of the form

$$\theta_{n+1} = \theta_n + \alpha_n\big[h(\theta_n) + w_n\big], \tag{18}$$

and guarantees $\theta_n \to \theta^*$ (the root of h) almost surely, provided:

(RM1) h has a unique root θ^* and is "well behaved" (e.g., $\langle \theta - \theta^*, h(\theta)\rangle < 0$ for $\theta \ne \theta^*$).

(RM2) The step sizes satisfy $\sum_n \alpha_n = \infty$ and $\sum_n \alpha_n^2 < \infty$.

(RM3) $\{w_n\}$ is a martingale difference sequence: $\mathbb{E}[w_n \mid \mathcal{F}_n] = 0$.

(RM4) The noise has bounded conditional variance: $\mathbb{E}[w_n^2 \mid \mathcal{F}_n] \le C(1 + \|\theta_n\|^2)$.

7.2 Applying the Template to TD(0)

Theorem 7.1 (Convergence of TD(0))**.** *Let $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ be a finite MDP. Let π be a stationary policy such that the induced Markov chain is ergodic. If the step sizes satisfy $\sum_t \alpha_t(s) = \infty$ and $\sum_t \alpha_t(s)^2 < \infty$ for every state s, and every state is visited infinitely often, then the TD(0) iterates satisfy*

$$V_t(s) \xrightarrow{a.s.} V^\pi(s) \qquad \textit{for all } s \in \mathcal{S}. \tag{19}$$

Proof sketch. We verify the four Robbins–Monro conditions:

1. (RM1): The mean field $h(V)(s) = (\mathcal{T}^\pi V)(s) - V(s)$ has unique root V^π (the fixed point of $\mathcal{T}^\pi$). Furthermore, $\mathcal{T}^\pi$ is a γ-contraction, so $\|h(V) + V - V^\pi\|_\infty = \|\mathcal{T}^\pi V - V^\pi\|_\infty \leq \gamma\|V - V^\pi\|_\infty$, which implies the required stability condition.
2. (RM2): This is a hypothesis of the theorem.
3. (RM3): Theorem 5.1, Part 1.
4. (RM4): Theorem 5.1, Part 2.

The convergence then follows from the Robbins–Monro theorem (Lesson 31, Theorem 2.1) or, equivalently, from the ODE method (Lesson 32) applied to the limiting ODE $\dot{v} = \mathcal{T}^\pi v - v$ whose globally asymptotically stable equilibrium is V^π. □

The Architecture of RL Convergence Proofs

Every convergence proof for a tabular RL algorithm follows the same three-step architecture:

1. **Rewrite** the update in Robbins–Monro form: identify $h(\theta)$ and w_n.
2. **Verify the noise conditions** using the TD noise decomposition (this lesson): show that w_n is an MDS with bounded conditional variance.
3. **Verify the mean-field conditions:** show that h has a unique root and satisfies a stability/contraction condition.

The probabilistic content is entirely in Step 2. Steps 1 and 3 are algebraic and analytic, respectively.

8 Summary of Key Results (Lessons 1–14)

The following table collects every key result from the Measure Theory track and explains its role in RL theory. Results are organized by lesson and type.

Result (Lesson)	**Statement**	**Role in RL**
Foundations (Lessons 1–5)		
Sigma-algebra axioms (L1)	Closed under complement and countable union	Event space of an MDP
Measure properties (L2)	Countable additivity, continuity from below/above	Probability axioms for $\mathbb{P}$
Measurable functions (L3)	Preimage of Borel sets is measurable	Random variables, policies as measurable maps
Linearity of $\mathbb{E}$ (L5)	$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$	Splitting $G_t = R_{t+1} + \gamma G_{t+1}$
MCT / DCT (L5)	Monotone and dominated convergence	Interchanging limits and $\mathbb{E}$, infinite-horizon value functions
Conditioning and Independence (Lessons 6–7)		
Cond. expectation (L6)	$\mathbb{E}[X\|\mathcal{G}]$ as L^2 projection	Foundation of Bellman equation derivation
Tower property (L6)	$\mathbb{E}[\mathbb{E}[X\|\mathcal{G}]] = \mathbb{E}[X]$	Bellman recursion, MDS $\Rightarrow$ zero mean
Pull-out property (L6)	$\mathbb{E}[h(Z)X\|Z] = h(Z)\mathbb{E}[X\|Z]$	Extracting $V(s)$ from $\mathbb{E}[\delta_t\|S_t]$
Total variance (L7)	$\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X\|Y)] + \mathrm{Var}(\mathbb{E}[X\|Y])$	Bias-variance tradeoff in baselines
Independence (L7)	$\mathbb{E}[\prod X_i] = \prod \mathbb{E}[X_i]$	Variance of sums, noise bounds
Martingales and Filtrations (Lesson 8)		
MDS definition (L8)	$\mathbb{E}[w_n\|\mathcal{F}_{n-1}] = 0$	All SA convergence proofs
MDS uncorrelated (L8)	$\mathbb{E}[w_m w_n] = 0$ for $m \neq n$	Noise variance bounds in SA
Inequalities (Lesson 9)		
Markov (L9)	$\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$	Foundation of all tail bounds
Chebyshev (L9)	$\mathbb{P}(\|X - \mu\| \geq \varepsilon) \leq \sigma^2/\varepsilon^2$	Weak law of large numbers
Hoeffding (L9)	Sub-Gaussian tail for bounded averages	Finite-sample MC bounds
Azuma–Hoeffding (L9)	Concentration for bounded MDS	Finite-sample TD bounds
Jensen (L9)	$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)]$	$\|\mathbf{P}_\pi \mathbf{v}\|_{d^\pi}$ contraction
Convergence and Limits (Lessons 10–11)		
Convergence modes (L10)	a.s. $\Rightarrow$ prob. $\Rightarrow$ dist.; $L^p \Rightarrow$ prob.	Interpreting convergence theorems
Borel–Cantelli (L10)	$\sum \mathbb{P}(A_n) < \infty \Rightarrow \mathbb{P}(A_n \text{ i.o.}) = 0$	Upgrading to a.s. convergence
Strong LLN	$\bar{X} \xrightarrow{\text{a.s.}}$	MC value estimation

The Complete Pipeline

The table above traces a complete logical chain from the axioms of probability to the convergence of TD(0):

$$\underbrace{\text{Sigma-algebras}}_{\text{L1–2}} \to \underbrace{\mathbb{E},\ \mathbb{E}[\cdot|\cdot]}_{\text{L5–6}} \to \underbrace{\text{MDS}}_{\text{L8}} \to \underbrace{\text{Convergence}}_{\text{L10–11}} \to \underbrace{\text{Ergodic thm}}_{\text{L12}} \to \underbrace{\text{Robbins–Siegmund}}_{\text{L13}} \to \underbrace{\text{TD no}}_{\text{L14}}$$

Each arrow represents a logical dependency: every result on the right uses results from the left.

9 Exercises

Exercise 9.1 (SARSA Noise Decomposition (∗))**.** In SARSA, the update uses the actual next action A_{t+1}:

$$Q_{t+1}(S_t, A_t) = Q_t(S_t, A_t) + \alpha_t\big[R_{t+1} + \gamma Q_t(S_{t+1}, A_{t+1}) - Q_t(S_t, A_t)\big].$$

Define the SARSA Bellman operator $(\mathcal{T}^\pi Q)(s,a) = \sum_{s'} P(s'|s,a)[r(s,a,s') + \gamma \sum_{a'} \pi(a'|s')Q(s',a')]$ and the noise w_t^{SARSA}.

(a) Write w_t^{SARSA} explicitly.

(b) Prove that $\mathbb{E}[w_t^{\text{SARSA}} \mid \mathcal{F}_t] = 0$.

(c) Show that $\mathbb{E}[(w_t^{\text{SARSA}})^2 \mid \mathcal{F}_t] \le C(1 + \|Q_t\|_\infty^2)$.

Exercise 9.2 (Non-Markovian Noise (∗∗))**.** Consider a POMDP where the agent observes $O_t = g(S_t)$ instead of the true state S_t, and maintains value estimates $V_t(o)$ for observations o.

(a) Define the analogous noise term $w_t(o)$ for a TD-like update on observations.

(b) Show by counterexample (a simple 3-state POMDP with 2 observations) that $\mathbb{E}[w_t(o) \mid \mathcal{F}_t] \neq 0$ in general.

(c) Explain intuitively why the Markov property is essential for the MDS cancellation.

Exercise 9.3 (Variance Reduction via Baselines (∗∗))**.** Consider the modified TD noise with a state-dependent baseline $b(s)$:

$$\tilde{w}_t(s) = R_{t+1} + \gamma V_t(S_{t+1}) - b(s).$$

(a) For what choice of $b(s)$ is $\tilde{w}_t(s)$ a martingale difference sequence?

(b) Show that the conditional variance is minimized (over deterministic baselines b) when $b(s) = (\mathcal{T}^\pi V_t)(s)$, which recovers the original noise $w_t(s)$.

(c) Use the total variance formula (Lesson 7) to decompose $\mathrm{Var}(\tilde{w}_t(s) \mid \mathcal{F}_t)$ into contributions from action randomness and transition randomness.

Exercise 9.4 (Noise in n-Step TD (∗∗∗))**.** In n-step TD, the update target is $G_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k R_{t+k+1} + \gamma^n V_t(S_{t+n})$. Define the noise $w_t^{(n)} = G_t^{(n)} - (\mathcal{T}^\pi)^n V_t(S_t)$.

(a) Prove that $\mathbb{E}[w_t^{(n)} \mid \mathcal{F}_t] = 0$. *Hint:* Use induction on n and the tower property.

(b) Show that $\mathbb{E}[(w_t^{(n)})^2 \mid \mathcal{F}_t] \le C_n(1 + \|V_t\|_\infty^2)$ where C_n grows with n.

(c) Discuss the bias-variance tradeoff: why does increasing n reduce bias (since $(\mathcal{T}^\pi)^n V \to V^\pi$) but increase variance (since C_n grows)?

Exercise 9.5 (Verifying Robbins–Monro Conditions $(*)$)**.** For the Q-learning update (14) in a finite MDP with $\gamma = 0.9$ and $R_{\max} = 1$:

(a) Write the update in Robbins–Monro form $Q_{t+1}(s,a) = Q_t(s,a) + \alpha_t[h(Q_t)(s,a) + w_t^Q(s,a)]$. Identify h explicitly.

(b) Compute the constant C' from Proposition 6.1 for these parameters.

(c) Verify that $h(Q)(s,a) = (\mathcal{T}^* Q)(s,a) - Q(s,a) = 0$ if and only if $Q = Q^*$ (the optimal action-value function).

10 Summary

This lesson established the central probabilistic result connecting measure theory to reinforcement learning:

- The **TD noise decomposition** (Theorem 5.1) shows that the noise in temporal-difference updates is a martingale difference sequence with bounded conditional variance. The proof uses two ingredients: the Markov property (to collapse conditioning on the full history to the current state-action pair) and reward boundedness (to control the variance).
- The same decomposition extends to **Q-learning** (Proposition 6.1) and **expected SARSA**, with essentially identical proofs.
- Combined with the **Robbins–Monro framework** (Lessons 31–32), the noise decomposition yields the convergence of TD(0), Q-learning, and SARSA under standard step-size conditions.
- The **comprehensive summary table** (Section 8) traces the logical dependencies from sigma-algebras (Lesson 1) through the TD noise decomposition (this lesson), showing how every measure-theoretic result contributes to the convergence proof pipeline.

This completes the Measure Theory track. The tools developed in Lessons 1–14 will be used repeatedly in Phases 2–4 whenever we prove convergence of an RL algorithm.

11 Further Reading

- **Stochastic approximation theory:** Borkar [2008] provides a comprehensive treatment of the Robbins–Monro and ODE methods with applications to RL. See especially Chapters 2–3 for the noise conditions and Chapter 5 for two-timescale SA.
- **RL convergence proofs:** Szepesvári [2010], Chapter 4, presents detailed convergence proofs for TD(0) and Q-learning with explicit noise decomposition. Bertsekas [2012], Chapter 6, covers the same material from a dynamic programming perspective.
- **Martingale theory background:** Williams [1991] is an excellent introduction to the probability theory underlying this lesson. Durrett [2019], Chapters 4–5, provides a more advanced treatment of martingale convergence.
- **Next steps:** Lesson 15 begins the Functional Analysis track, which provides the operator-theoretic framework (contraction mappings, projections) that complements the probabilistic tools from this track. Lessons 31–32 in the Optimization track formalize the Robbins–Monro and ODE methods that consume the noise conditions proved here.

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